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Modélisation, Analyse Numérique et Calcul Scientifique

Supervised Study Project in Mathematics(SSPM) Report

Shallow water equations

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Introduction

The study of geophysical fluid dynamics has become increasingly important in the context of climate change, particularly due to the growing need to understand and predict phenomena such as sea level rise, coastal flooding, and tsunami propagation. These processes involve complex fluid flows whose accurate modelling is essential for both environmental risk assessment and engineering applications.

The shallow water equations (SWE) arise as a simplified model of incompressible fluid dynamics derived from the Euler equations under the assumption of a hydrostatic pressure distribution and a small aspect ratio between the vertical and horizontal length scales. In this regime, vertical accelerations are neglected compared to horizontal motion, leading to a depth-averaged description of the flow. This reduction results in a system of nonlinear hyperbolic conservation laws governing the evolution of the water height and the depth-averaged velocity.

From a mathematical point of view, the SWE constitute a fundamental model in the theory of hyperbolic systems, exhibiting rich phenomena such as wave propagation, shock formation, and rarefaction waves. Their analysis combines tools from partial differential equations, conservation laws, and numerical analysis.

The objectives of this project are therefore:

1. To understand the model.
2. To study its mathematical properties.
3. to develop a numerical scheme for its resolution using Fortran 90,
4. and to validate the model on representative test cases.

To this end, we progressively construct the model starting from the linear transport equation, then Burgers' equation as a nonlinear prototype, before arriving at the full shallow water system.

$$\begin{cases} \partial_t h + \partial_x (hu) = 0 \\ \partial_t (hu) + \partial_x \left(hu^2 + \dfrac{gh^2}{2} \right) = 0 \end{cases}$$

Chapter 1

1D constant-velocity transport equation – linear case

Considering the case of a wave, the phenomenon can be simplified as the transport, displacement of a water height from point a to point b, similar to a vehicle moving in uniform rectilinear motion.

We know that transport equations are given by:

Let $a \in \mathbb{R}$, $(t, x) \in \mathbb{R}_+ \times \mathbb{R}$ and $w \in C^1(\mathbb{R})$,

$$\partial_t w + \partial_x f(w) = 0$$

with:

- t : time variable
- x : spatial variable
- w : transported quantity
- f : flux

In the linear case, we obtain the following problem:

$$\text{(PC)} \quad \begin{cases} \partial_t w + a\,\partial_x w = 0 \\ w(t=0, x) = w_0(x) \end{cases}$$

1.1 Analytical solution

The linear advection equation belongs to the class of first-order hyperbolic partial differential equations. For such equations, information propagates along characteristic curves defined by the underlying transport velocity.

The method of characteristics consists in rewriting the PDE as an ordinary differential equation along these curves. This transformation is justified by the chain rule, which

shows that the solution satisfies a conservation property along trajectories satisfying $\frac{dx}{dt} = a$.

This approach reduces the original PDE to a simple ODE, making it possible to obtain an explicit representation of the solution.

Thus using the method of characteristics to solve (PC), we seek curves $x(t)$ in the (x, t) plane such that the solution remains constant along these curves, i.e.:

$$\frac{d}{dt}w(x(t), t) = 0$$

By the chain rule:

$$\frac{d}{dt}w(x(t), t) = \partial_t w + \frac{dx}{dt}\partial_x w$$

Comparing with the equation:

$$\partial_t w + a\partial_x w = 0$$

we identify:

$$\frac{dx}{dt} = a$$

We obtain an ODE whose solution is:

$$x(t) = x_0 + at$$

Thus, characteristics are straight lines in the (x, t) plane.

Moreover, $\frac{dw}{dt} = 0$, hence:

$$w(x(t), t) = w(x_0, 0)$$

Since $x(t) = x_0 + at$, we have $x_0 = x - at$.

In conclusion, the solutions of (PC) are:

$$w(x, t) = w_0(x - at)$$

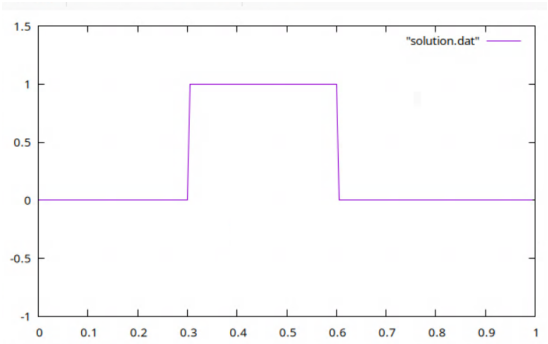
1.2 Numerical solution

Using the Lax-Friedrichs scheme and initializing the following parameters:

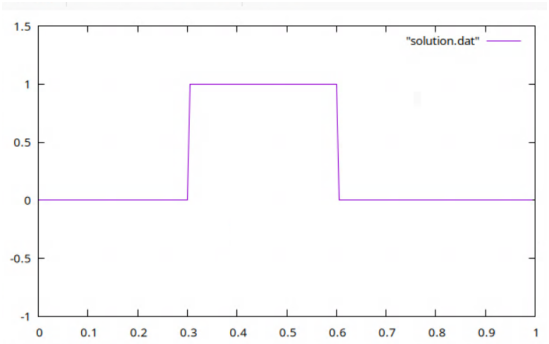
$$a = 1, \quad CFL = \frac{1}{2}, \quad L = 1 \text{ i.e } (x \in [0, 1]), \quad T = 1 \text{ i.e } (t \in [0, 1])$$

we obtain the following curves for

$$w_0(x) = \begin{cases} 1 & \text{if } 0.3 < x < 0.6 \\ 0 & \text{sinon} \end{cases}$$



(a) initial solution



(b) solution with Lax-friedrichs, CFL = 0.5

Interpretation The graph shows the propagation of the square wave under the advection equation. The numerical solution obtained with the Lax–Friedrichs scheme becomes progressively smoother, illustrating the numerical diffusion introduced by the method. While the wave is correctly transported, the discontinuities are no longer sharp compared to the exact solution.

Chapter 2

Burgers' equation: transport equation – nonlinear case

In the linear case, the medium transports the quantity without being influenced by it; the system is therefore passive. Example: all cars move at constant speed, so there is no traffic congestion. However, in reality, this is not the case. In the nonlinear case, the transport is self-influenced; the phenomenon generates its own flux, leading to internal interactions.

In the car example, the velocity depends on density: the more cars there are, the slower the traffic becomes, leading to shock formation (traffic jams).

For Burgers' equation,

$$f(w) = \frac{w^2}{2}$$

We then obtain the following PDE:

$$\begin{cases} \partial_t w + \partial_x \left(\dfrac{w^2}{2} \right) = 0 \\ w(x, t = 0) = w_0(x) \end{cases}$$

2.1 Analytical solution

We know that velocity is not constant,

$$\text{velocity} = w(x(t), t)$$

Using the method of characteristics:

$$\begin{cases} \dfrac{dx}{dt} = w(x(t), t) \\ x(t = 0) = x_0 \end{cases}$$

We introduce:

$$\varphi(t) = w(x(t), t)$$

and differentiating:

$$\frac{d\varphi}{dt} = \frac{dx(t)}{dt}\partial_x w(x(t), t) + \partial_t w(x(t), t)$$

$$= w(x(t), t)\partial_x w(x(t), t) + \partial_t w(x(t), t)$$

$$= (\partial_t w + w\partial_x w)(x(t), t)$$

$$= 0$$

Thus, the solution remains constant along characteristics.
Since φ is constant along characteristics:

$$\frac{dx}{dt} = w_0(x_0)$$

whose solution is:

$$x(t) = x_0 + w_0(x_0)t$$

We know that:

1. $w(x, t) = w_0(x_0)$
2. $x = x_0 + w_0(x_0)t$

We invert:

$$x = x_0 + w_0(x_0)t$$

to recover:

$$w(x, t) = w_0(x_0)$$

Thus:

1. If the function
$$x = F(x_0) = x_0 + w_0(x_0)t$$
is strictly increasing, then it is invertible.

$$F'(x_0) = 1 + t\, w_0'(x_0) > 0$$

therefore

$$w(x,t) = w_0(F^{-1}(x))$$

and

$$x_0 = F^{-1}(x)$$

is unique, so the solution is called "regular".

2. If the function is not strictly increasing:

$$F'(x_0) = 1 + t\, w_0'(x_0) \leq 0$$

then characteristics intersect, leading to a shock.

NB: The solution is no longer physically meaningful because it becomes multi-valued.

Thus $\exists x_0 < x_1$ such that $w_0(x_0) = w_0(x_1)$

In the presence of a shock, the solution becomes discontinuous.

Note : Let's introduce the anotation : w_L : the expression of $w(x,t)$ on the left of the discontinuity and w_R, on the right.

- We then seek a weak solution.
- By the Rankine–Hugoniot condition:

$$\sigma(w_R - w_L) = f(w_R) - f(w_L)$$

This indicates that the shock speed is:

$$\sigma = \frac{f(w_R) - f(w_L)}{w_R - w_L}$$

- The shock appears at time $T^* > 0$:

$$T^* = -\frac{1}{\min\limits_{x \in \mathbb{R}} \dfrac{d}{dx} f'(w_0(x))}$$

In conclusion, if w_0 is strictly increasing, the solution is regular for all $t > 0$. Otherwise, it becomes discontinuous for $t > T^*$.

2.2 Numerical solution

The chosen scheme is the Lax-Friedrichs scheme:

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x}\left(F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}\right)$$

with flux:

$$F_{i+\frac{1}{2}} = \frac{1}{2}\left(f(w_i^n) + f(w_{i+1}^n)\right) - \frac{\Delta x}{2\Delta t}\left(w_{i+1}^n - w_i^n\right)$$

CFL condition:

$$\frac{\Delta t}{\Delta x}\max|f'(w)| \leq \frac{1}{2}$$

and since

$$f(w) = \left(\frac{1}{2}w^2\right)' = w$$

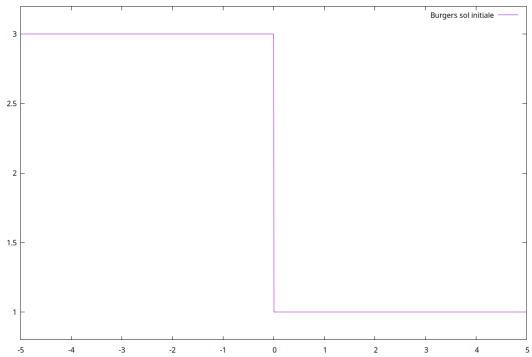
we obtain

$$\frac{\Delta t}{\Delta x}\max|w_i^n| \leq \frac{1}{2}$$

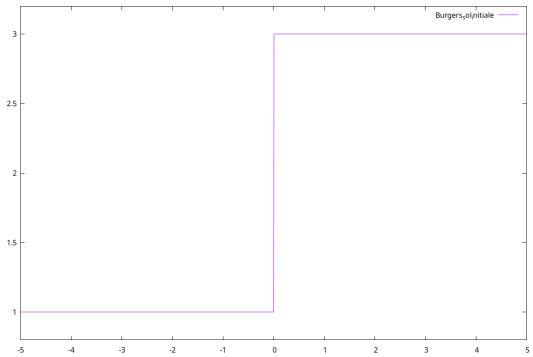
Hence:

$$\Delta t = 0.5\frac{\Delta x}{\max|w_i^n|}$$

Initial solution illustration



(a) CFL = 0.5, wR = 1, wL = 3, C=1



(b) CFL = 0.5, wR = 3, wL = 1, C=1

Chapter 3

The Riemann problem – for Burgers' equation

We observe that Burgers' equations show that in the nonlinear case, starting from:

- an irregular initial condition ($w_0(x)$),
- non-constant velocity,

characteristics intersect, leading to a shock at time T^*. A discontinuity is therefore generated by the dynamics.

With the Riemann problem, we study the evolution of this discontinuity in time: what becomes of it, and in which direction it propagates.

Let:

$$\begin{cases} \partial_t w + \partial_x f(w) = 0 \\ w(x, t=0) = \begin{cases} w_L & \text{if } x < 0 \\ w_R & \text{if } x > 0 \end{cases} \end{cases}$$

There are multiple solutions because of discontinuities; the equation is no longer satisfied in the classical sense.

A possible solution is:

$$-\sigma(w_R - w_L) = f(w_R) - f(w_L)$$

$$\Rightarrow \quad \sigma = \frac{f(w_R) - f(w_L)}{w_R - w_L}$$

shock speed.

Global conservation is not sufficient to ensure uniqueness. Since the equation is not satisfied classically, multiple weak solutions exist, which are mathematically meaningful but physically non-unique.

3.1 Study of multiple solutions – Example

Let w be a function with a discontinuity point.

$$\begin{cases} \partial_t w + \partial_x \dfrac{w^2}{2} = 0 \\ w(x, t = 0) = w_0 \end{cases}$$

with:

$$w(x,t) = \begin{cases} 0 & \text{if } x < -at \\ -2a & \text{if } -at < x < 0 \\ 2a & \text{if } 0 < x < at \\ 0 & \text{if } x > at \end{cases}$$

Using Rankine–Hugoniot to verify weak solution property:

- 1st discontinuity:

$$\sigma = -a, \qquad w_L = 0, \qquad w_R = -2a$$

then:

$$-\sigma(w_R - w_L) + \left(\frac{w_R^2}{2} - \frac{w_L^2}{2}\right)$$

$$= a((-2a) - 0) + \left(\frac{(-2a)^2}{2} - \frac{0^2}{2}\right)$$

$$= -2a^2 + 2a^2$$

$$= 0$$

- 2nd discontinuity:

$$\sigma = 0, \qquad w_L = -2a, \qquad w_R = 2a$$

$$= 0$$

- 3rd discontinuity:

$$\sigma = a, \qquad w_L = 2a, \qquad w_R = 0$$

$$= 0$$

Conclusion: we obtain 3 weak solutions, mathematically acceptable but physically ambiguous, so an additional selection principle is needed.

3.2 Selection of physical solutions

Let

$$\partial_t w + \partial_x f(w) = 0 (\star)$$

Consider a viscous problem:

$$\partial_t w^\varepsilon + \partial_x f(w^\varepsilon) + \varepsilon \partial_{xx} w^\varepsilon = 0$$

with

$$\varepsilon \to 0$$

Let $\eta \in C^2$ convex.

Multiplying $(\star)$ by $\eta'(w^\varepsilon)$ gives:

$$\eta'(w^\varepsilon)\partial_t w^\varepsilon + \eta'(w^\varepsilon)\partial_x f(w^\varepsilon) = \varepsilon \eta'(w^\varepsilon)\partial_{xx} w^\varepsilon$$

We define:

$$\eta'(w^\varepsilon)\partial_x f(w^\varepsilon) = \partial_x G(w^\varepsilon)$$

and

$$\eta'(w^\varepsilon)\partial_x w^\varepsilon = \varepsilon \left(\eta'(w^\varepsilon)\partial_{xx} w^\varepsilon\right)$$

and:

$$\eta'(w^\varepsilon)\partial_x w^\varepsilon = \partial_t(\eta'(w^\varepsilon)\partial_x w^\varepsilon) - \eta''(w^\varepsilon)(\partial_x w^\varepsilon)^2$$

Thus:

$$\partial_t \eta(w^\varepsilon) + \partial_x G(w^\varepsilon) = \varepsilon \partial_x \left(\eta'(w^\varepsilon)\partial_x w^\varepsilon\right)$$

By convexity of η:

$$-\eta''(w^\varepsilon)(\partial_x w^\varepsilon)^2 \leq 0$$

Then we pass to the limit $\varepsilon \to 0$:

$$\partial_t \eta(w) + \partial_x G(w) \leq 0$$

This is the entropy condition selecting the physical solution.

Therefore, under entropy conditions, C^1 functions with discontinuities satisfy:

- Rankine–Hugoniot

$$-\sigma(w_R - w_L) + (f(w_R) - f(w_L)) = 0$$

- Entropy condition

$$-\sigma(\eta(w_R) - \eta(w_L)) + (G(w_R) - G(w_L)) \leq 0$$

Example

For Burgers' equation:

$$\eta : w \mapsto \frac{w^2}{2}$$

with

$$G' = \eta' f'$$

thus $G'(w) = w.w = w^2$

Hence

$$G(w) = \frac{w^3}{3}$$

Counter-example

$\sigma = 0$, $w_L = -2a$, $w_R = 2a$

Entropy condition gives:

$$= \frac{16}{3}a^3 > 0$$

Thus $w(x,t)$ is not an entropy weak solution.

Remark:

For Burgers' equation, an entropy shock means:

$$-\sigma(\eta(w_R) - \eta(w_L)) + (G(w_R) - G(w_L)) = 0$$

For all convex η and $G' = \eta' f' \iff w_L \geq w_R$

3.3 Riemann problem solution – Burgers

$$\begin{cases} \partial_t w + \partial_x f(w) = 0 \\ w(x, t=0) = \begin{cases} w_L & \text{if } x < 0 \\ w_R & \text{if } x > 0 \end{cases} \end{cases}$$

- Case 1 : $w_L > w_R$
  The solution is a shock (discontinuity) propagating with speed

$$\sigma = \frac{f(w_R) - f(w_L)}{w_R - w_L}$$

$$w(x,t) = \begin{cases} w_L & \text{if } x < \sigma t \\ w_R & \text{if } x > \sigma t \end{cases}$$

- Case 2 : $w_L < w_R$
  The solution is continuous.

$$w(x,t) = \begin{cases} w_L & \text{if } x < f'(w_L)t \\ ? & \\ w_R & \text{if } x > f'(w_R)t \end{cases}$$

  The solution is of the form $w(x,t) = v(\frac{x}{t})$ [admitted]
  Injecting into the PDE gives:

$$\partial_t w(x,t) = -\frac{x}{t^2} v'(\frac{x}{t})$$

  and

$$\partial_x w(x,t) = \frac{1}{t} v'(\frac{x}{t})$$

  Thus:

$$\left(-\frac{x}{t} + f'(v(\frac{x}{t}))\right) v'(\frac{x}{t}) = 0$$

  Hence either $v' = 0$ or $f'(v) = x/t$.
  The first case is impossible since it would not connect w_L and w_R.
  Thus $f'(v(\frac{x}{t})) = \frac{x}{t}$; let $a(w) = f'(w)$.

$$v(\frac{x}{t}) = a^{-1}(\frac{x}{t})$$

  Therefore:

$$w(x,t) = \begin{cases} w_L & \text{if } x < w_L t \\ \frac{x}{t} & \text{if } w_L t < x < w_R t \\ w_R & \text{if } x > w_R t \end{cases}$$

* For Burgers:

$$f(w) = \frac{w^2}{2}$$

$$a(w) = f'(w) = w$$

$$a^{-1}(\frac{x}{t}) = \frac{x}{t}$$

$$w(x,t) = \begin{cases} w_L & \text{if } x < w_L t \\ \frac{x}{t} & \text{if } w_L t < x < w_R t \\ w_R & \text{if } x > w_R t \end{cases}$$

Conclusion of the Riemann problem solution:

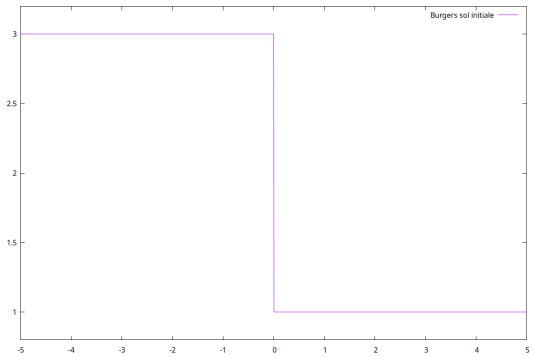
We have seen that weak solutions satisfy Rankine–Hugoniot but not always entropy. Therefore the entropy condition selects the physically relevant weak solution.

Entropy gives a simple rule:

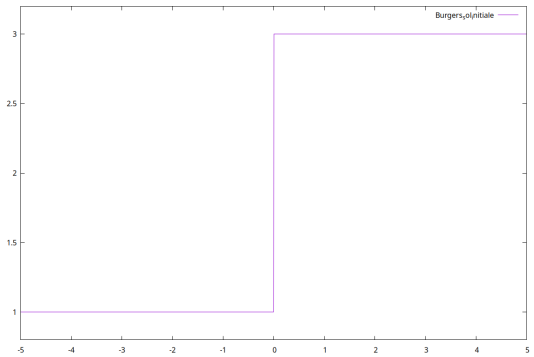
- $w_L > w_R \Rightarrow$ shock
- $w_L < w_R \Rightarrow$ rarefaction

3.4 Numerical solution.

Initial solution illustration

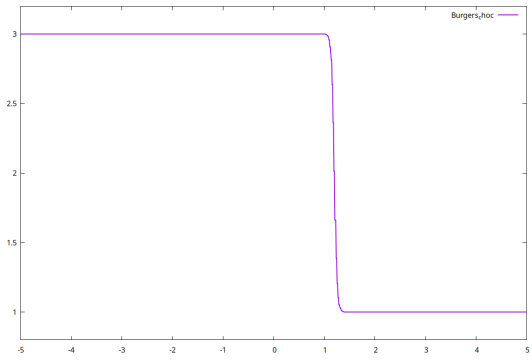


(a) CFL = 0.5, wR = 1, wL = 3, C=1

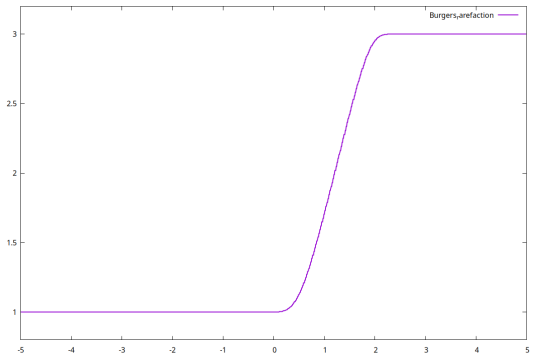


(b) CFL = 0.5, wR = 3, wL = 1, C=1

Solution illustration



(a) CFL = 0.5, wR = 1, wL = 3, C=1



(b) CFL = 0.5, wR = 3, wL = 1, C=1

Interpretation The graphs show the time evolution of the Riemann problem for Burgers' equation, where an initial discontinuity separates two constant states. Depending on the sign of w_L and w_R, the solution develops either a shock wave (compressive case) or a rarefaction fan (expansive case). In the shock case, a sharp discontinuity propagates at constant speed given by the Rankine–Hugoniot condition, while in the rarefaction case the solution spreads smoothly between the two states. The numerical results capture this self-similar structure, but the shock is slightly smeared due to numerical diffusion. Overall, the simulations are consistent with the expected entropy solution of the Riemann problem.

Chapter 4

Shallow Water Equations

In this final chapter, we study the shallow water equations. They constitute a natural generalization of the models previously investigated in this work, namely the transport equation and Burgers' equation. After studying wave propagation phenomena and the formation of discontinuities in the scalar case, the introduction of the shallow water equations allows us to consider the more general framework of coupled hyperbolic systems.

4.1 Formulation of the Equation

The study of wave motion in shallow water leads to a system of **conservation laws**. We examine the motion of water (a fluid) in a channel, simplifying the problem by focusing on the water height $h(x,t)$ and assuming that its vertical velocity is negligible compared to its horizontal velocity $v(x,t)$. This assumption is valid for small-amplitude waves in a fluid where the depth is small compared to the wavelength.

Since we consider only the horizontal direction x, these equations are **one-dimensional**: variations in width or depth are ignored. Consequently, all quantities depend only on x (position) and t (time).

Conservation Laws

The system can be written in the general form:

$$\partial_t w + \partial_x f(w) = 0,$$

where:

$$w = \begin{pmatrix} h \\ q \end{pmatrix}, \quad f(w) = \begin{pmatrix} q \\ \frac{q^2}{h} + \frac{gh^2}{2} \end{pmatrix}, \quad q = hu.$$

Conservation of Mass

$$\partial_t h + \partial_x (hu) = 0.$$

Here:

- h is the water height (amount of water).
- hu is the water flow.

Interpretation: The amount of water changes only if water enters or exits the system (i.e., "what exits = what enters").

Conservation of Momentum

$$\partial_t(hu) + \partial_x\left(hu^2 + \frac{1}{2}gh^2\right) = 0.$$

Here:

- hu is the momentum.
- $\frac{1}{2}gh^2$ is the water pressure due to gravity.

Interpretation: The speed of the water changes due to motion and gravity.

Conservative Form

In order to facilitate both the theoretical analysis and the numerical discretization, the system is rewritten in conservative form:

$$\partial_t U + \partial_x F(U) = 0.$$

with

$$U = \begin{pmatrix} h \\ q \end{pmatrix} \qquad \text{and} \qquad F(U) = \begin{pmatrix} q \\ \dfrac{q^2}{h} + \dfrac{1}{2}gh^2 \end{pmatrix}.$$

This formulation highlights the conservation law structure of the system and allows the use of numerical methods specifically designed for hyperbolic systems.

4.2 Riemann Problem

As in the case of Burgers' equation, the shallow water equations may develop non-smooth solutions involving shock waves or rarefaction waves.
The Riemann problem consists in considering a discontinuous initial condition of the form:

$$U(x, t = 0) = \begin{cases} U_L & \text{if } x < 0 \\ U_R & \text{if } x > 0 \end{cases}$$

The study of this problem is essential in the framework of finite volume methods. Indeed, most modern numerical schemes rely on the local resolution of Riemann problems at each interface between computational cells.

In this work, the tools previously developed for Burgers' equation served as the foundation for the study of the shallow water system.

Remark

Unlike the linear transport equation and Burgers' equation, the shallow water equations form a nonlinear coupled system of conservation laws. As a consequence, explicit analytical solutions are generally not available in the general case.

However, in some simplified configurations, analytical solutions can still be derived. For instance, the Riemann problem with piecewise constant initial data admits self-similar solutions involving shock waves and rarefaction waves. Similarly, linearized versions of the shallow water equations around a constant steady state lead to explicit wave-like solutions.

In the general nonlinear setting considered in this work (in particular for the dam-break configuration), the coupling between the variables and the nonlinear structure of the flux generate complex wave interactions and possible discontinuities. These features make the derivation of closed-form solutions intractable in practice.

For this reason, numerical methods are essential tools for approximating the solutions of the shallow water equations.

4.3 Numerical Resolution

4.3.1 Lax-Friedrichs Scheme

The numerical resolution of the shallow water equations was carried out using the Lax–Friedrichs scheme. This explicit scheme is written as:

$$U_i^{n+1} = \frac{1}{2}\left(U_{i+1}^n + U_{i-1}^n\right) - \frac{\Delta t}{2\Delta x}\left(F(U_{i+1}^n) - F(U_{i-1}^n)\right)$$

The Lax–Friedrichs scheme has the advantage of being simple to implement while ensuring good numerical stability for hyperbolic systems.

However, it introduces relatively strong numerical diffusion, particularly noticeable near discontinuities.

4.3.2 CFL stability Condition

When solving hyperbolic partial differential equations using explicit numerical schemes, the **CFL (Courant-Friedrichs-Lewy) condition** acts as a fundamental requirement for numerical stability.
In a simple linear case (such as the 1D advection equation, in which motion moves at a constant speed c), the CFL condition states that the distance traveled by the physical wave during one time-step Δt must not exceed the spatial mesh size Δx. This is expressed through the dimensionless Courant number C:

$$C = c\frac{\Delta t}{\Delta x} \leq 1 \tag{4.1}$$

If $C > 1$, the physical domain of dependence lies outside the numerical domain of dependence, meaning information propagates faster than the grid can capture, leading to catastrophic numerical expansion (instability).

For the non-linear Shallow Water Equations, the constant velocity c must be replaced.

Mathematical Derivation of the Wave Speeds ($\lambda_{\max}$)

To determine the maximum speed at which information propagates through the system—which is crucial for establishing the CFL stability condition—we must analyze the eigenvalues of the system. This is achieved by transforming the conservative system into its quasi-linear form.

Step 1: The Quasi-Linear Form and the Jacobian Matrix

The standard compact form of a system of conservation laws is given by:

$$\partial_t w + \partial_x f(w) = 0 \tag{4.2}$$

Using the chain rule for differentiation, the spatial derivative of the flux vector can be expanded as $\partial_x f(w) = \frac{\partial f}{\partial w}\partial_x w$. This allows us to rewrite the system in a quasi-linear form:

$$\partial_t w + A(w)\partial_x w = 0 \tag{4.3}$$

where $A(w) = \frac{\partial f}{\partial w}$ is the **Jacobian matrix** of the flux function. Given the state vector w and the flux vector $f(w)$:

$$w = \begin{pmatrix} h \\ q \end{pmatrix}, \quad f(w) = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} q \\ \frac{q^2}{h} + \frac{1}{2}gh^2 \end{pmatrix}$$

The Jacobian matrix $A(w)$ is a 2×2 matrix defined by the partial derivatives of f with respect to the components of w (h and q):

$$A(w) = \begin{pmatrix} \frac{\partial f_1}{\partial h} & \frac{\partial f_1}{\partial q} \\ \frac{\partial f_2}{\partial h} & \frac{\partial f_2}{\partial q} \end{pmatrix} \tag{4.4}$$

Step 2: Computing the Components of $A(w)$

We evaluate each partial derivative independently, substituting the physical velocity $u = q/h$ where appropriate:

- $\frac{\partial f_1}{\partial h} = \frac{\partial}{\partial h}(q) = 0$
- $\frac{\partial f_1}{\partial q} = \frac{\partial}{\partial q}(q) = 1$
- $\frac{\partial f_2}{\partial h} = \frac{\partial}{\partial h}\left(\frac{q^2}{h} + \frac{1}{2}gh^2\right) = -\frac{q^2}{h^2} + gh = -u^2 + gh$
- $\frac{\partial f_2}{\partial q} = \frac{\partial}{\partial q}\left(\frac{q^2}{h} + \frac{1}{2}gh^2\right) = \frac{2q}{h} = 2u$

Assembling these components, the Jacobian matrix expressed in terms of the physical variables h and u is:

$$A(w) = \begin{pmatrix} 0 & 1 \\ -u^2 + gh & 2u \end{pmatrix} \tag{4.5}$$

Step 3: Finding the Eigenvalues

The characteristic speeds of propagation correspond to the eigenvalues λ of the Jacobian matrix $A(w)$. These are found by solving the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix:

$$\det \begin{pmatrix} -\lambda & 1 \\ -u^2 + gh & 2u - \lambda \end{pmatrix} = 0$$

Computing the determinant yields:

$$(-\lambda)(2u - \lambda) - 1 \times (-u^2 + gh) = 0$$

$$\lambda^2 - 2u\lambda + u^2 - gh = 0$$

This quadratic equation can be simplified by recognizing a perfect square in the first three terms:

$$(\lambda - u)^2 - gh = 0 \implies (\lambda - u)^2 = gh$$

Taking the square root on both sides provides the two distinct eigenvalues of the system:

$$\lambda_1 = u - \sqrt{gh}, \quad \lambda_2 = u + \sqrt{gh} \tag{4.6}$$

Physical Interpretation and $\lambda_{\max}$

The two eigenvalues represent the physical speeds at which waves travel through the channel:

- $\lambda_1 = u - \sqrt{gh}$ is the speed of an upstream-propagating wave (moving against the flow).
- $\lambda_2 = u + \sqrt{gh}$ is the speed of a downstream-propagating wave (moving with the flow).

Here, $\sqrt{gh}$ represents the natural speed of waves at the surface. For the numerical simulation to remain stable (CFL criterion), the time-step Δt must be calculated based on the fastest wave speed in the entire channel. We therefore choose the maximum absolute value between our two eigenvalues:

$$\lambda_{\max} = \max(|\lambda_1|, |\lambda_2|) = |u| + \sqrt{gh} \tag{4.7}$$

Conclusion: The Non-Linear CFL Condition

Since the fluid velocity u_i and water height h_i vary at each grid node i, the maximum wave speed $\lambda_{\max}$ changes dynamically across the domain. Consequently, to guarantee stability everywhere in our simulation, the global Courant number C for the Shallow Water system is defined as:

$$C = \max_i \left(|u_i| + \sqrt{g h_i} \right) \frac{\Delta t}{\Delta x} \leq 1 \tag{4.8}$$

Which give us :

$$\Delta t \leq \frac{\Delta x}{\max_i \left(|u_i| + \sqrt{g h_i} \right)} \tag{4.9}$$

4.3.3 Riemann Solver

The principle of finite volume methods relies on the computation of numerical fluxes at the interfaces between cells. At each interface, the states coming from neighboring cells locally define a Riemann problem. The Riemann solver developed in the previous sections was therefore used to construct the numerical fluxes required for the discretization of the system.

In the case of the shallow water equations, the left and right states are given by:

$$U_L = \begin{pmatrix} h_L \\ q_L \end{pmatrix}, \qquad U_R = \begin{pmatrix} h_R \\ q_R \end{pmatrix}.$$

The Riemann solver then provides an approximation of the numerical flux at the interface while taking into account the propagation of the characteristic waves of the system.

This approach generalizes the methods previously used for Burgers' equation to the case of hyperbolic systems of conservation laws.

4.4 Dam-break problem (Riemann test case)

To validate the numerical scheme, we consider the classical dam-break problem, which is a standard benchmark for the shallow water equations. This configuration corresponds to a Riemann-type initial condition and allows us to test the ability of the numerical method to capture nonlinear wave propagation, including shock waves and rarefaction waves.

The initial condition consists of two constant states separated by an interface:

$$U(x, t = 0) = \begin{cases} U_L & \text{if } x < x_0 \\ U_R & \text{if } x > x_0 \end{cases}$$

where $U_L = (h_L, q_L)^T$ represents a higher water level (reservoir) and $U_R = (h_R, q_R)^T$ corresponds to a lower water level (dry or nearly dry region).

This configuration models the sudden removal of a barrier separating two fluid regions, leading to the propagation of nonlinear waves.

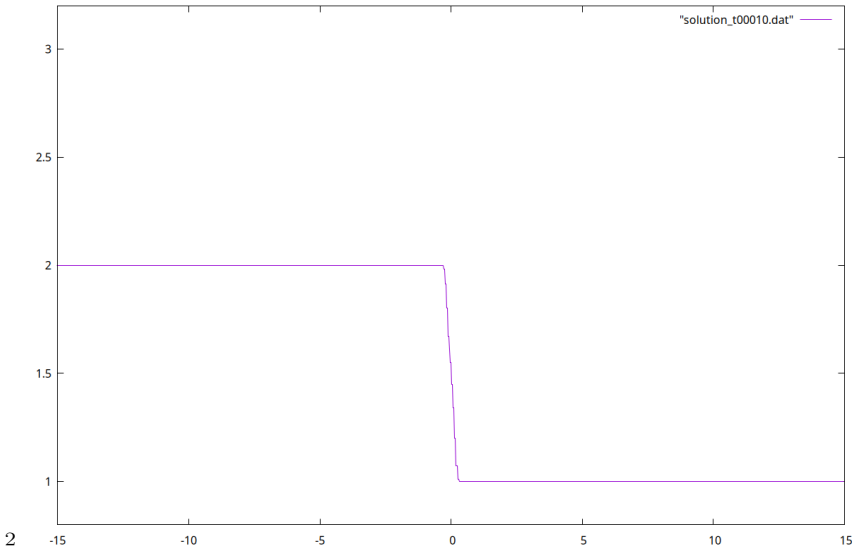
Expected dynamics

Depending on the initial height difference, the solution typically consists of:

- a rarefaction wave propagating into the deeper region,
- a shock wave propagating into the shallower region,
- and a contact between both states governed by the characteristic structure of the system.

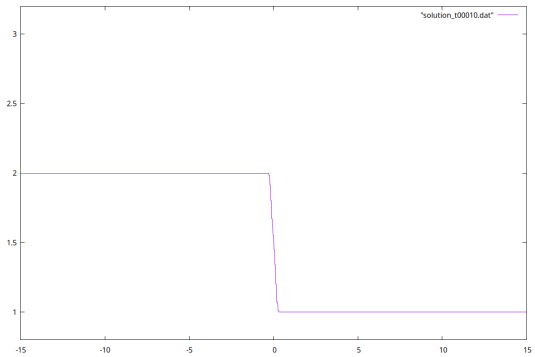
Numerical results

The dam-break problem is simulated using the Lax–Friedrichs scheme introduced previously. The results show the formation and propagation of nonlinear waves, which are consistent with the theoretical structure of the shallow water equations.

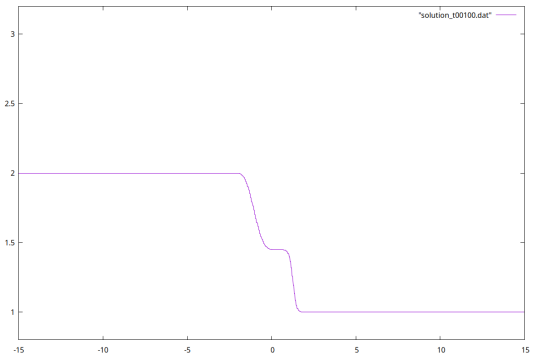


solution illustration

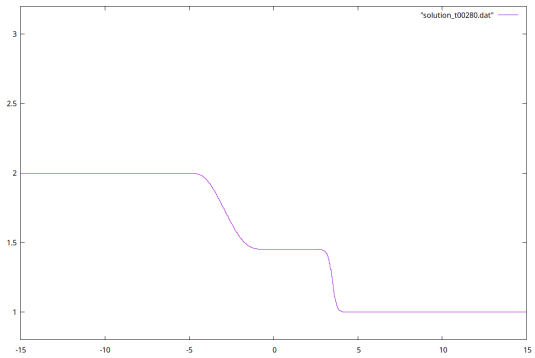
Dam-break simulation using the Lax–Friedrichs scheme :



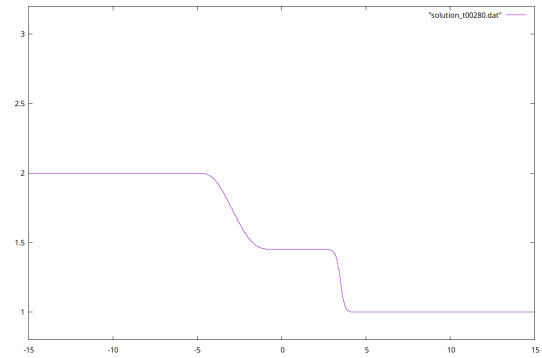
(a) step 1



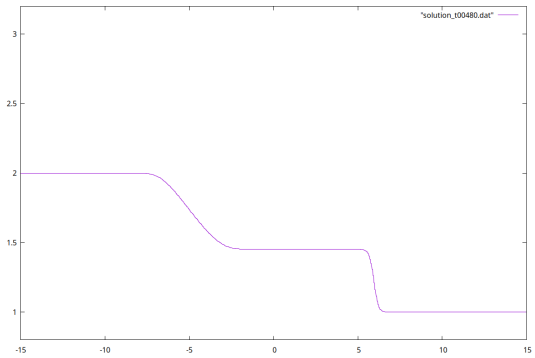
(b) step 100



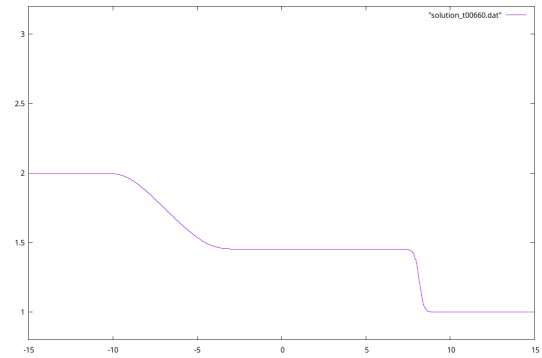
(c) step 280



(d) step 320



(e) step 480



(f) step 660

Interpretaion The dam-break simulation illustrates the nonlinear wave structure of the shallow water equations resulting from a Riemann initial condition. After the removal of the initial discontinuity, the solution separates into two distinct wave patterns governed by the characteristic speeds $u \pm \sqrt{gh}$

On the side of higher water depth, a rarefaction wave develops, leading to a smooth decrease in water height and an acceleration of the flow. This expansion fan spreads progressively in time, consistent with the hyperbolic nature of the system.

On the opposite side, a fast-moving front propagates into the shallow region, corresponding to a shock-like wave associated with the abrupt increase in water height. This discontinuity travels at a speed determined by the Rankine–Hugoniot condition and is well captured qualitatively by the numerical scheme.

Between these two structures, the flow evolves toward a smoother intermediate state where momentum and mass are redistributed. The numerical results are consistent with

the expected Riemann structure of the shallow water equations.

Finally, the simulations highlight the effect of numerical diffusion: the Lax–Friedrichs scheme tends to smear the shock and wetting front, while still preserving the global wave structure.

4.4.1 Limitations of the Lax–Friedrichs scheme

Although the Lax–Friedrichs scheme is simple to implement and robust for hyperbolic conservation laws, it presents several important limitations that affect its accuracy and practical use.

First, the scheme is only first-order accurate in both space and time. As a consequence, it introduces significant numerical diffusion, which becomes particularly visible near discontinuities such as shock waves or contact discontinuities. These sharp structures are artificially smeared over several grid cells, reducing the sharpness of the solution.

Second, the method is not based on an exact or approximate Riemann solver at cell interfaces. Instead, it relies on a strong averaging procedure combined with a diffusive term. While this contributes to its stability, it also leads to excessive dissipation of physically relevant features, especially in long-time simulations.

Third, the numerical viscosity introduced by the scheme is independent of the physical properties of the system. This may result in an unphysical smoothing of waves and a loss of small-scale structures, which are essential in nonlinear hyperbolic systems such as the shallow water equations.

For these reasons, although the Lax–Friedrichs scheme is well-suited for qualitative simulations and validation purposes, it is not sufficient when accurate resolution of discontinuities or high-resolution wave structures is required. This motivates the use of higher-order reconstruction techniques such as MUSCL.

4.4.2 Second-Order Reconstruction: The MUSCL Scheme

The Limitation of First-Order Schemes In the Finite Volume Method, the computational domain is divided into discrete cells. By default, standard first-order numerical schemes (such as the basic Lax-Friedrichs scheme) assume that the solution is constant within each cell. This is known as a piecewise constant approximation.

Consequently, when calculating the flux at the interface between two cells, the scheme directly uses the cell-centered average values.

While this approach is highly stable, it introduces a significant amount of numerical dissipation, causing excessive diffusion that artificially flattens sharp fronts such as the shock wave in a dam-break simulation.

The MUSCL Approach To mitigate numerical diffusion and achieve second-order spatial accuracy, the MUSCL (Monotonic Upstream-centered Scheme for Conservation Laws) approach is implemented. Originally introduced by van Leer, MUSCL replaces the piecewise constant assumption with a piecewise linear reconstruction.

Instead of assuming the water level is flat inside the cell, MUSCL calculates a local slope (σ_i) for the conserved variables based on neighboring cells. This allows the solver to extrapolate the exact states of the fluid immediately at the left and right boundaries of the cell interface, rather than relying on the distant cell-center averages.

Mathematically, for a cell i, the reconstructed states at the right interface $(i + 1/2)$ and the left interface $(i - 1/2)$ are defined as:

$$w^L_{i+1/2} = w_i + \frac{1}{2}\sigma_i \tag{4.10}$$

$$w^R_{i-1/2} = w_i - \frac{1}{2}\sigma_i \tag{4.11}$$

These accurately reconstructed boundary states (w^L and w^R) are then fed directly into the Lax-Friedrichs numerical scheme to compute the flux at the interface.

Slope Limiters While the piecewise linear reconstruction dramatically improves the sharpness of shock waves, it can introduce unphysical oscillations near steep gradients (a phenomenon linked to Godunov's theorem). To preserve monotonicity and prevent the creation of artificial extrema (such as negative water heights), a slope limiter is applied to σ_i.

In this project, the **Minmod** (Min-Modulus) limiter is used. It compares the backward difference $(w_i - w_{i-1})$ and the forward difference $(w_{i+1} - w_i)$ across the cell. The limited slope σ_i is computed as:

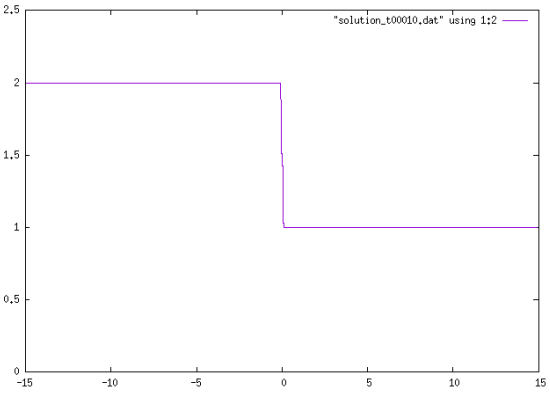
$$\sigma_i = \text{minmod}(w_i - w_{i-1}, w_{i+1} - w_i) \tag{4.12}$$

The standard *minmod* function is defined as:

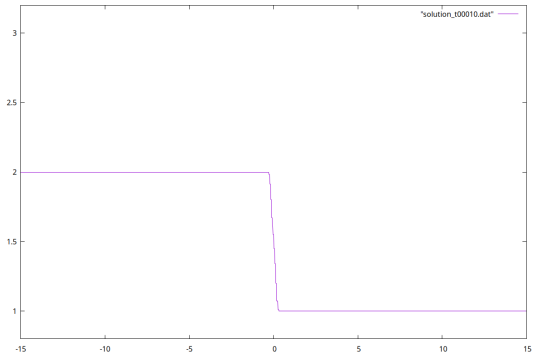
$$\text{minmod}(a, b) = \begin{cases} \min(a, b) & \text{if } a > 0 \text{ and } b > 0 \\ \max(a, b) & \text{if } a < 0 \text{ and } b < 0 \\ 0 & \text{if } a \cdot b \leq 0 \end{cases} \tag{4.13}$$

Logically, this function returns the smallest slope in absolute value if both gradients have the same sign. If the signs differ, it returns zero. This mechanism effectively reduces the scheme to first-order accuracy in the immediate vicinity of severe discontinuities, ensuring stability while preserving second-order accuracy in smooth regions.

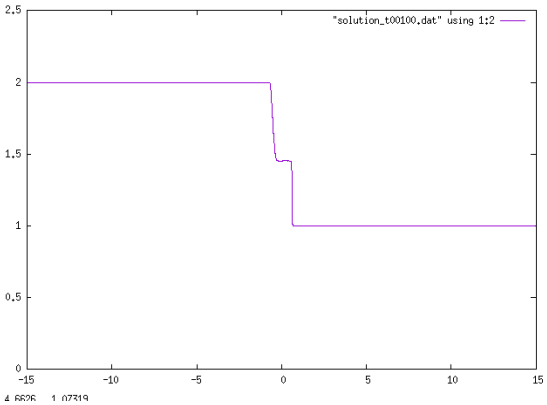
Comparison between Lax-Friedrichs and MUSCL reconstruction



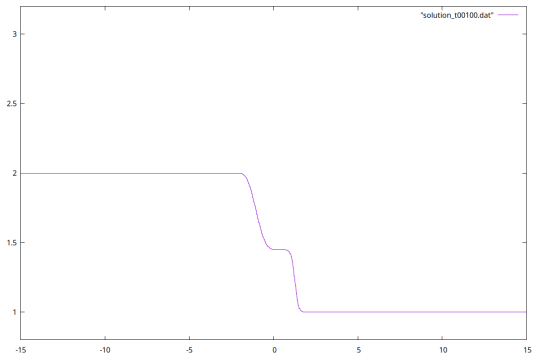
(a) MUSCL solution step 10



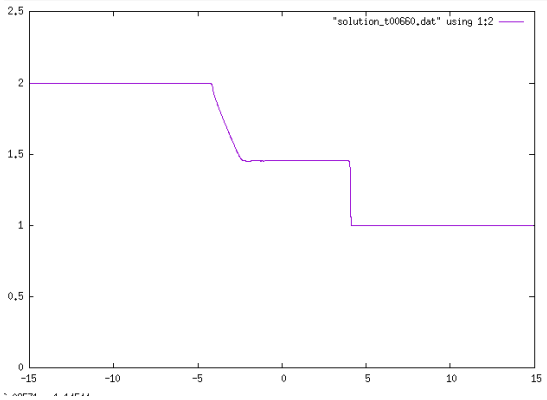
(b) Lax-friedrichs solution step 10



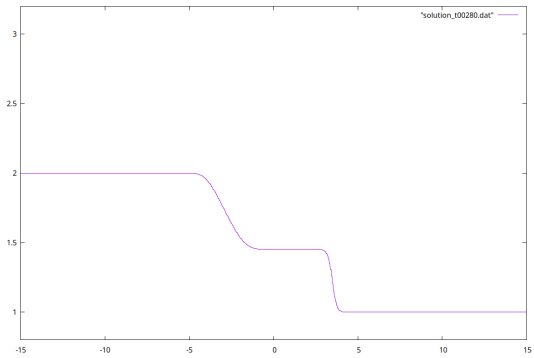
(c) MUSCL solution step 100



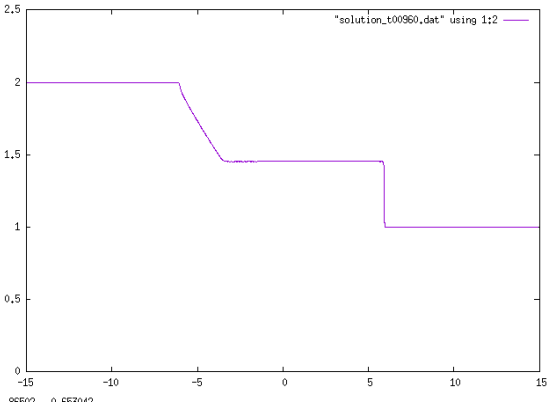
(d) Lax-friedrichs solution step 100



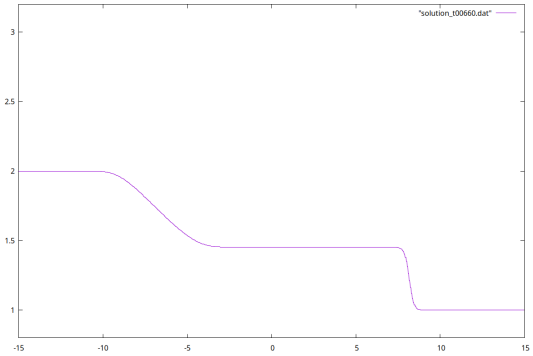
(e) MUSCL solution step 280



(f) Lax-friedrichs solution step 280



(g) MUSCL solution step 660



(h) Lax-friedrichs solution step 660

Interpretaion

The following analysis evaluates the performance of the first-order **Lax-Friedrichs** scheme against the second-order **MUSCL** reconstruction, applied to the Shallow Water Equations (SWE). The results are illustrated at several time steps: $t = 10$, $t = 100$, $t = 280$, and $t = 660$.

1. Observations on the Lax-Friedrichs Scheme

The first-order approach is characterized by significant **numerical diffusion**. As the simulation progresses, this manifests through:

- **Severe Smoothing:** The sharp shock front is significantly degraded, causing the physical discontinuity to spread over a wide range of grid points.
- **Dissipative Error:** The scheme fails to preserve the sharpness of the hydraulic jump, leading to a progressive loss of physical accuracy as t increases.

2. Benefits of the MUSCL Reconstruction

By incorporating MUSCL reconstruction, the scheme significantly enhances the resolution of discontinuities:

- **Reduced Dissipation:** The second-order reconstruction effectively counteracts numerical diffusion, allowing the scheme to maintain a much steeper and physically accurate shock front.
- **Improved Gradient Capture:** The expansion wave and the jump conditions remain well-defined and sharp, even at $t = 660$.

In summary, this demonstrates the efficiency of second-order reconstruction in resolving sharp discontinuities in the Shallow Water Equations.

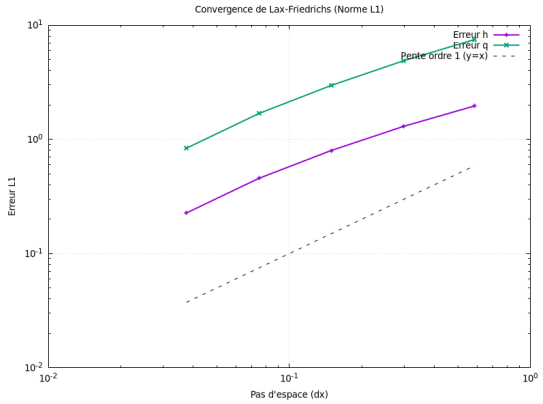
Numerical Analysis of the errors in norm L1

Theory states that the Lax-Friedrichs method is a first-order spatial and temporal scheme. Mathematically, this implies that if the spatial step Δx is divided by two (by doubling the number of grid points N_s), the numerical error should also be divided by two.

We can verify this behavior using our obtained results:

- **From $N_s = 400$ to $N_s = 800$:** The error decreases from 0.455 to 0.226. Dividing 0.455 by 2 yields 0.2275, which aligns almost perfectly with the theoretical factor of 2, confirming a first-order convergence rate.
- **From $N_s = 200$ to $N_s = 400$:** The error decreases from 0.798 to 0.455, resulting in a ratio of approximately 1.75, which consistently approaches the first-order asymptotic behavior as the grid refines.

On large grids (such as $N_s = 50$ or $N_s = 100$), the error ratio is not exactly 2. This happens because the Lax-Friedrichs scheme is naturally diffusive: it adds "artificial viscosity" to keep the simulation stable. Instead of capturing a sharp wave, it blurs the shock across several grid cells, which creates a large initial error. However, as the number of points increases, the wave becomes sharper and the method perfectly matches its theoretical first-order behavior.



(a) Order 1

```
Calcul de la solution de reference (Ns =        3200 )...
Test avec Ns =          50
  Erreur L1 h:   1.9630362361044078
Test avec Ns =         100
  Erreur L1 h:   1.2959917943099730
Test avec Ns =         200
  Erreur L1 h:  0.79844888620208854
Test avec Ns =         400
  Erreur L1 h:  0.45595113780176538
Test avec Ns =         800
  Erreur L1 h:  0.22695945432591400
Etude terminee ! Fichier 'convergence_LF.dat' cree.
```

(b) Values

Graph Analysis

The figure above shows the L_1 norm error as a function of the spatial step Δx on a logarithmic scale. Analyzing this plot confirms how the scheme behaves:

- **Convergence Slope:** The error curves for both water height (h) and discharge (q) are perfectly parallel to the black dashed line (which has a slope of 1). On a log-log plot, this parallel behavior proves experimentally that the Lax-Friedrichs scheme is strictly first-order accurate.
- **Error Reduction:** We observe that the error decreases steadily as the mesh size Δx decreases. Even though the error is significant on coarse meshes, because the method smooths out abrupt changes, it follows the expected slope of 1 very closely as more points are added. This confirms that the solver is stable and correct.
- **Conclusion:** The graph clearly demonstrates that our code follows the mathematical theory of the Lax-Friedrichs method. The constant slope across all tested grid sizes proves that the numerical implementation works reliably.

Conclusion

In this work, we have developed a progressive study of hyperbolic conservation laws, starting from the linear transport equation, then Burgers' equation, and finally the shallow water system. This approach allowed us to progressively introduce key mathematical concepts such as characteristics, weak solutions, entropy conditions, and nonlinear wave interactions.

The shallow water equations were analyzed in conservative form, and their characteristic structure was derived through the eigenvalues of the Jacobian matrix. This analysis led naturally to the CFL stability condition, which governs the numerical discretization.

A finite volume framework based on the Lax–Friedrichs scheme was implemented and validated through classical test cases, including the dam-break problem. The numerical results correctly reproduce the expected wave structure, consisting of rarefaction and shock waves, in agreement with the Riemann problem theory.

However, the Lax–Friedrichs scheme exhibits significant numerical diffusion, which limits its ability to accurately capture sharp discontinuities. To address this limitation, a MUSCL reconstruction with slope limiting was introduced, significantly improving the resolution of steep gradients while maintaining stability.

Overall, this study provides a consistent introduction to the numerical approximation of nonlinear hyperbolic systems. Extensions of this work could include higher-order Godunov-type methods, more accurate Riemann solvers, and multidimensional shallow water simulations for realistic geophysical applications.

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